arXiv:2410.05229v1 [cs.LG] 7 Oct 2024

GSM-Symbolic: Understanding the Limitations of Mathematical Reasoning in Large Language Models

Iman Mirzadeh† Keivan Alizadeh Hooman Shahrokhi*
Oncel Tuzel Samy Bengio Mehrdad Farajtabar†

Apple

Abstract

Recent advancements in Large Language Models (LLMs) have sparked interest in their formal reasoning capabilities, particularly in mathematics. The GSM8K benchmark is widely used to assess the mathematical reasoning of models on grade-school-level questions. While the performance of LLMs on GSM8K has significantly improved in recent years, it remains unclear whether their mathematical reasoning capabilities have genuinely advanced, raising questions about the reliability of the reported metrics. To address these concerns, we conduct a large-scale study on several state-of-the-art open and closed models. To overcome the limitations of existing evaluations, we introduce GSM-Symbolic, an improved benchmark created from symbolic templates that allow for the generation of a diverse set of questions. GSM-Symbolic enables more controllable evaluations, providing key insights and more reliable metrics for measuring the reasoning capabilities of models.Our findings reveal that LLMs exhibit noticeable variance when responding to different instantiations of the same question. Specifically, the performance of all models declines when only the numerical values in the question are altered in the GSM-Symbolic benchmark. Furthermore, we investigate the fragility of mathematical reasoning in these models and demonstrate that their performance significantly deteriorates as the number of clauses in a question increases. We hypothesize that this decline is due to the fact that current LLMs are not capable of genuine logical reasoning; instead, they attempt to replicate the reasoning steps observed in their training data. When we add a single clause that appears relevant to the question, we observe significant performance drops (up to 65%) across all state-of-the-art models, even though the added clause does not contribute to the reasoning chain needed to reach the final answer. Overall, our work provides a more nuanced understanding of LLMs' capabilities and limitations in mathematical reasoning.

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities across various domains, including natural language processing, question answering, and creative tasks (Gunter et al., 2024; OpenAI, 2023; Dubey et al., 2024; Anil et al., 2023; Abdin et al., 2024; Rivière et al., 2024). Their potential to perform complex reasoning tasks, particularly in coding and mathematics, has garnered significant attention from researchers and practitioners.

However, the question of whether current LLMs are genuinely capable of true logical reasoning remains an important research focus. While some studies highlight impressive capabilities, a closer examination reveals substantial limitations. Literature suggests that the reasoning process in LLMs

*Work done during an internship at Apple. †Correspondence to {imirzadeh,farajtabar}@apple.com.

GSM8K

When Sophie watches her nephew, she gets out a variety of toys for him. The bag of building blocks has 31 blocks in it. The bin of stuffed animals has 8 stuffed animals inside. The tower of stacking rings has 9 multicolored rings on it.Sophie recently bought a tube of bouncy balls, bringing her total number of toys for her nephew up to 62. How many bouncy balls came in the tube?

Let T be the number of bouncy balls in the tube.
After buying the tube of balls, Sophie has 31+8+9+ T = 48 + T =62 toys for her nephew.
Thus, T =62-48 = <<62-48=14>>14 bouncy balls came in the tube.

GSM Symbolic Template

When {name} watches her {family}, she gets out a variety of toys for him. The bag of building blocks has {x} blocks in it. The bin of stuffed animals has {y} stuffed animals inside.The tower of stacking rings has {z} multicolored rings on it.{name} recently bought a tube of bouncy balls, bringing her total number of toys she bought for her {family} up to {total}. How many bouncy balls came in the tube?

```
#variables:
- name = sample(names)
- family = sample(["nephew", "cousin", "brother"])
- x = range(5, 100)
- y = range(5, 100)
- z = range(5, 100)
- total = range(100, 500)
- ans = range(85, 200)

#conditions:
- x + y + z + ans == total
```

Let T be the number of bouncy balls in the tube. After buying the tube of balls, {name} has {x} + {y} + {z} + T = { x + y + z } + T = {total} toys for her {family}.

Thus, T = {total} - { x + y + z } = <<{total}-{ x + y + z }={ans}>>{ans} bouncy balls came in the tube.

Figure 1: Illustration of the `GSM-Symbolic` template creation process. This dataset serves as a tool to investigate the presumed reasoning capabilities of LLMs, enabling the design of controllable mathematical reasoning evaluations with more reliable metrics. Our results reveal that all state-of-the-art LLMs exhibit significant performance variations, suggesting the fragility or lack of reasoning.

is probabilistic pattern-matching rather than formal reasoning (Jiang et al., 2024). Although LLMs can match more abstract reasoning patterns, they fall short of true logical reasoning. Small changes in input tokens can drastically alter model outputs, indicating a strong token bias and suggesting that these models are highly sensitive and fragile (Jiang et al., 2024; Shi et al., 2023). Additionally, in tasks requiring the correct selection of multiple tokens, the probability of arriving at an accurate answer decreases exponentially with the number of tokens or steps involved, underscoring their inherent unreliability in complex reasoning scenarios (Schaeffer et al., 2023).

Mathematical reasoning is a crucial cognitive skill that supports problem-solving in numerous scientific and practical applications. Consequently, the ability of large language models (LLMs) to effectively perform mathematical reasoning tasks is key to advancing artificial intelligence and its real-world applications. The GSM8K (Grade School Math 8K) dataset (Cobbe et al., 2021) has emerged as a popular benchmark for evaluating the mathematical reasoning capabilities of LLMs. While it includes simple math questions with detailed solutions, making it suitable for techniques like Chain-of-Thought (CoT) prompting, it provides only a single metric on a fixed set of questions. This limitation restricts comprehensive insights into the models' mathematical reasoning. Moreover, the popularity and prevalence of GSM8K can increase the risk of inadvertent data contamination. Finally, the static nature of GSM8K does not allow for controllable experiments to understand model limitations, such as behavior under varied conditions or changes in question aspects and difficulty levels.

To address these limitations, a more versatile and adaptive evaluation framework is needed—one that can generate diverse question variants and adjust complexity levels to better explore the robustness and reasoning abilities of LLMs. This would facilitate a deeper understanding of the strengths and weaknesses of these models in mathematical reasoning tasks. We make the following contributions:

- We introduce `GSM-Symbolic`, an enhanced benchmark that generates diverse variants of `GSM8K` questions using symbolic templates (Sec. 3), as shown in Fig. 1. This enables a more nuanced and reliable evaluation of LLMs' performance across various setups, moving beyond single-point accuracy metrics. Our large-scale study on 25 state-of-the-art open and closed models provides significant insights into LLMs' behavior in mathematical reasoning tasks.

- We question the *reliability* of currently reported results on `GSM8K` and demonstrate that the performance of LLMs can be viewed as a distribution with unwarranted variance across different instantiations of the same question. We show that the performance of all models drops on `GSM-Symbolic` (Sec. 4.1), hinting at potential data contamination.

- We show that LLMs exhibit more robustness to changes in superficial elements like proper names but are very sensitive to changes in numerical values (Sec. 4.2). We show that performance degradation and variance increase as the number of clauses increases, indicating that LLMs' reasoning capabilities struggle with increased complexity (Sec. 4.3).

- Finally, we further question the reasoning abilities of LLMs and introduce the `GSM-NoOp` dataset. By adding seemingly relevant but ultimately irrelevant information to problems, we demonstrate substantial performance drops (up to 65%) across all state-of-the-art models (Sec. 4.4). This reveals a critical flaw in the models' ability to discern relevant information for problem-solving, likely because their reasoning is not formal in the common sense term and is mostly based on pattern matching. We show that even when provided with multiple examples of the same question or examples containing similar irrelevant information, LLMs struggle to overcome the challenges posed by `GSM-NoOp`. This suggests deeper issues in their reasoning processes that cannot be alleviated by in-context shots and needs further investigation.

Overall, our work provides a comprehensive understanding of the limitations of LLMs in mathematical reasoning. Our results emphasize the need for more reliable evaluation methodologies and further research into the reasoning capabilities of large language models.

2 Related Work: Reasoning & Language Models

Logical reasoning is a critical trait of intelligent systems. Recent advancements in Large Language Models (LLMs) have demonstrated significant potential across various domains, yet their reasoning abilities remain uncertain and inconsistent. Many works have investigated whether LLMs are truly capable of reasoning by examining how these models solve tasks requiring logical reasoning.
One interesting direction focuses on modeling the computation performed by transformers. For example, parallels have been drawn between components such as attention and feed-forward modules and simple computational primitives (Weiss et al., 2021; Zhou et al., 2024). Delétang et al. (2023) demonstrated that transformers fail to generalize on non-regular tasks and showed that structured memory (e.g., memory tape) is necessary for handling complex tasks. This is related to the effectiveness of Chain-of-Thought (CoT) prompting (Wei et al., 2022) and using scratchpads for LLMs as additional memory for intermediate computations. Overall, current results suggest that while the transformer architecture has limitations and lacks the required expressiveness for solving